

MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images

Sukhrobbek Ilyosbekov
Northeastern University
Portland, ME, USA
ilyosbekov.s@northeastern.edu

Shubham Gajjar
Northeastern University
Portland, ME, USA
gajjar.shu@northeastern.edu

Rongfei Jin
Northeastern University
Portland, ME, USA
jin.ron@northeastern.edu

Abstract—Cell Painting images record how cells respond to a chemical or genetic perturbation, and matching an image back to the perturbation that produced it is a basic step in phenotypic drug discovery that current methods still do poorly. We present MorphoCLIP, a contrastive model that places Cell Painting wells and text descriptions of perturbations in a shared 512-dimensional space. Each fluorescence channel passes through a frozen DINOv3 ViT-L/16, a one-layer CrossChannelFormer merges the five channel tokens, and a frozen BioClinical ModernBERT encodes the prompt; only the channel transformer and two projection heads are trained, about 14M parameters, and one model covers compounds, CRISPR knockouts and ORF overexpressions. With both backbones cached, an epoch over the CPJUMP1 pilot takes seconds on one 16GB GPU. Ranking held-out perturbation profiles against 98 candidate texts gives Recall@10 near 39% in both directions on the validation split (random: 10%), and 37% image-to-text and 44% text-to-image on the test split (random: 12%). We add three terms to the training signal, a replicate-alignment image loss, gene-aware soft labels and per-plate offsets measured against replicate plates, and measure each alone. Their effect on retrieval is within the spread a single checkpoint shows between splits at one seed. The replicate loss does raise replicability on the standard CPJUMP1 benchmark on all twelve tracks (mean fraction retrieved 0.18 to 0.25, and 0.28 with all three additions), a gain that is partly definitional; plate offsets alone leave the benchmark unchanged. On the same harness, compound replicability is level with or above the released CellCLIP checkpoint at one hundredth of its size, and compound target matching falls inside the published CellProfiler range. Cross-modality gene-compound matching is at or near zero for every variant (at most three of 179 to 480 compound-gene pairs pass the significance gate), and we report that as an open failure. We also document an evaluation pitfall: ranking wells against a text pool in which each text appears once per replicate well caps recall and made image-to-text retrieval look worse than chance.

Index Terms—contrastive learning, cell painting, morphological profiling, CLIP, perturbation matching

I. INTRODUCTION

Phenotypic drug discovery asks how a chemical or genetic perturbation changes the state of a cell, at a scale that target-specific assays cannot reach. The stakes are set by the cost of the alternative: bringing one drug to approval costs on the order of \$2B and takes about a decade [1], and roughly nine of ten compounds that enter clinical trials fail [2].

High-content microscopy is one way to get that scale. The Cell Painting assay [3], [4] stains cells with five fluorescent

dyes that mark DNA, endoplasmic reticulum, RNA and nucleoli, actin and Golgi, and mitochondria, and images them as a five-channel picture of cellular state. A perturbation changes that picture, and the change carries information about the perturbation’s mechanism. The CPJUMP1 benchmark [5] turns this into a retrieval problem over about 3 million fields of view: given images of cells treated with 303 compounds, 160 CRISPR knockouts and 176 ORF overexpressions, can a model match perturbations that act on the same target?

Existing methods each cover part of this problem, and none covers all of it. The CellProfiler pipeline [6] measures handcrafted features and recovers 5 to 25% of the expected sister-perturbation matches on CPJUMP1 [5]. CLOOME [7] and MolPhenix [8] learn contrastive image encoders against molecular fingerprints, and so handle compounds only. CellCLIP [9] replaces fingerprints with text prompts, which admits genetic perturbations, but runs on a 1.48B-parameter DINOv2-g stack and its cross-class retrieval trails CellProfiler. CWA-MSN [10] attacks batch effects directly and beats CellCLIP on gene-gene retrieval with 22M parameters, but has no text branch and cannot be queried in language.

MorphoCLIP puts text supervision, joint training over compounds and genes, and a batch correction step into one small model. It follows CLIP [11]: an image encoder and a text encoder are trained so that a Cell Painting well and the description of its perturbation land close together in a shared 512-dimensional space. Both backbones are frozen and their outputs are cached, so the trained part is a one-layer transformer over channel tokens plus two projection heads, about 14M parameters against CellCLIP’s 1.48B total.

The paper makes four contributions.

- 1) A text-supervised model over cached DINOv3 channel tokens [12], trained on compounds, CRISPR knockouts and ORF overexpressions together, that trains in seconds per epoch on one 16GB GPU once the cache is in memory.
- 2) Three additions to the training signal, each measured alone against a shared control: an image-image loss that pulls replicate wells of one perturbation together, gene-aware soft labels in the contrastive loss, and a per-plate offset measured against the replicate plates of the same condition. The replicate loss raises benchmark replicability on every track, and the three together raise

it further; on their own the other two are within single-seed noise.

- 3) An evaluation at the perturbation level with analytic random baselines, plus results on the standard CPJUMP1 benchmark. Compound replicability is level with or above the released CellCLIP checkpoint on the same harness, and compound target matching sits inside the published CellProfiler range. Cross-modality gene-compound matching is at or near zero for every variant.
- 4) A documented evaluation pitfall. Ranking wells against a candidate pool in which every text vector is duplicated once per replicate well caps recall near Recall@1 and breaks the shuffled random baseline. An earlier draft of this work concluded from that pool that image-to-text retrieval was below chance; on the corrected pool it is about four times chance.

II. PROBLEM STATEMENT

Given a Cell Painting image $\mathbf{x} \in \mathbb{R}^{5 \times H \times W}$ (five fluorescence channels, acquired at 1080×1080 pixels and resized to 384×384 for the backbone) and a text description t of the perturbation applied to the imaged cells, we learn encoders f_{img} and f_{txt} that map both into a shared unit sphere:

$$f_{\text{img}}(\mathbf{x}), f_{\text{txt}}(t) \in \mathbb{R}^d, \quad \|f_{\text{img}}(\mathbf{x})\|_2 = \|f_{\text{txt}}(t)\|_2 = 1. \quad (1)$$

Matching image-text pairs should have high cosine similarity and non-matching pairs low similarity. We use $d = 512$. In practice the image unit is a well: all sites of one well are encoded together and produce one vector.

The trained encoders support three retrieval tasks:

- **Text-to-image retrieval.** Given a perturbation description, retrieve the wells that show the corresponding phenotype.
- **Image-to-text retrieval.** Given a well, return the perturbation description that best matches its morphology.
- **Image-to-image retrieval.** Compare well embeddings by cosine similarity, without text at inference time. This is the setting of the standard CPJUMP1 benchmark: do replicate wells of one perturbation match, and do perturbations that share a target match across modalities?

Three properties of the data make this harder than natural image-caption alignment.

a) Batch effects.: The same perturbation looks different across plates and imaging days. Illumination, staining and focus drift add technical variance that is often larger than the biological signal. In CPJUMP1, replicability depends on modality (compounds replicate more cleanly than CRISPR) and on cell line (U2OS against A549) [5], and the distribution of per-perturbation average precision is heavily right-skewed: most perturbations score close to zero.

b) Fuzzy ground truth.: Perturbations do not come with clean positive and negative labels. Two compounds may share a primary target and diverge on secondary effects. A CRISPR knockout and a small molecule can converge on one pathway through different mechanisms. Similarity is a

continuum, which argues for soft-positive losses over hard-negative InfoNCE.

c) Scale and compute.: CPJUMP1 has 51 plates and 3.31 TiB of raw 16-bit TIFF images, about 3 million fields of view and 75 million cells. Training full vision and language encoders end to end at this scale is out of reach on one consumer GPU. We instead run the frozen backbones once, cache their outputs, and train only the small components that sit on top.

III. RELATED WORK

A. Morphological Profiling

Cell Painting [3] fixed the five-stain protocol that most high-content morphological profiling now uses [4]. The classical analysis tool, CellProfiler [6], measures more than 1,000 handcrafted features per cell. On CPJUMP1 [5] that pipeline retrieves 5 to 25% of the expected sister perturbations within a modality and 7 to 17% of CRISPR guide matches, and cross-modal compound-to-gene matching is harder still.

Handcrafted features can only measure what their authors thought to measure. Self-supervised vision transformers trained with DINO beat CellProfiler on drug-target and gene-family classification with zero-shot transfer [13], which is the case for building on frozen foundation-model features.

B. Contrastive Learning for Cell Painting

CLOOME [7] was the first CLIP-style model for Cell Painting. It trains a ResNet image encoder against a Morgan-fingerprint MLP and reaches top-1 retrieval more than $70 \times$ random over about 2,000 candidates. It handles chemical perturbations only and treats the five channels as RGB.

MolPhenix [8] swaps the trained image branch for a frozen Phenom-1 encoder, averages replicate embeddings, and adds a soft-to-label loss, gaining $8.78 \times$ over CLOOME. It also covers compounds only and depends on a proprietary backbone.

CellCLIP [9] is the closest prior work. It uses templated prompts (“A Cell Painting image of [cell_type] cells treated with [drug_name], SMILES: [SMILES_string]”), a Cross-ChannelFormer that encodes each channel separately before attending across channels, attention-based pooling across cells, and a Continuously Weighted Contrastive Loss (CWCL) for soft positives. It runs on DINOv2-g (1.1B parameters, 1.48B in total) and is evaluated on the Bray dataset [14] and on RxRx3-core gene-gene benchmarks (CORUM recall 0.354). Its text branch admits every perturbation type, but its cross-class mAP falls below the CellProfiler baseline, so text supervision alone does not solve the batch-effect problem.

C. Batch-Effect Correction

CWA-MSN [10] addresses batch effects inside a masked siamese network by aligning embeddings of cells that received the same perturbation across wells and plates. It absorbs technical variance without an explicit batch label. With 22M parameters and 0.2M training images it improves on CellCLIP by 9% and on OpenPhenom by 29% on gene-gene retrieval (CORUM recall 0.386). It has no text branch.

TABLE I

CELL PAINTING REPRESENTATION LEARNING METHODS. THE PARAMETER COUNT IS TRAINABLE PARAMETERS FOR MORPHOCLIP AND TOTAL PARAMETERS FOR THE OTHER LEARNED METHODS.

Method	Image Encoder	Text	Pert. Types	Batch Corr.	Params
CellProfiler [6]	Handcrafted	No	All	Sphering	N/A
CLOOME [7]	ResNet	No	Compounds	No	~25M
MolPhenix [8]	Phenom-1	No	Compounds	No	~36M
CellCLIP [9]	DINOv2-g	Yes	All	No	1,477M
CWA-MSN [10]	ViT-S/16	No	Unimodal	CWA	22M
MorphoCLIP	DINOv3-L	Yes	All	Plate offsets	~14M

D. Foundation Models for Microscopy

DINOv2 [15] and DINOv3 [12] give self-supervised visual features that transfer to biological imaging despite natural-image pretraining [13]. DINOv3 adds gram anchoring and a revised training recipe. Channel Vision Transformers [16] show that per-channel encoding followed by cross-channel attention works for multi-channel microscopy, which is the pattern our channel aggregator follows.

On the text side, BioClinical ModernBERT [17] (150M parameters) is pre-trained on 53.5B tokens of PubMed, PMC and clinical text, about $17\times$ the biomedical corpus behind the PubMedBERT (110M, 3.1B tokens) that CellCLIP uses. It is built on ModernBERT (rotary embeddings, Flash Attention, GeGLU) and can encode both small-molecule and gene descriptions with one model, unlike SMILES-specialized encoders such as ChemBERTa.

E. Positioning of MorphoCLIP

Table I places MorphoCLIP among these methods. CellCLIP has text but no batch correction. CWA-MSN has batch correction but no text. MorphoCLIP has both, trains on all three perturbation types, and keeps the trained part small. Where a like-for-like number exists we compare on the same benchmark harness (Section VI); elsewhere we quote published figures and say what differs.

IV. METHOD

MorphoCLIP has two encoders and one contrastive objective. The image encoder turns a well of five-channel Cell Painting sites into a 512-dimensional unit vector. The text encoder does the same for a prompt that describes the perturbation. Figure 1 shows the pipeline.

A. Image Encoder

A Cell Painting site has five fluorescence channels (Table II). We encode each channel on its own: the single-channel image is resized to 384×384 , copied to three channels, and passed through a frozen DINOv3 ViT-L/16 [12] (300M parameters). The result is five 1024-dimensional CLS tokens per site,

$$\{\mathbf{z}_c\}_{c=1}^5, \quad \mathbf{z}_c \in \mathbb{R}^{1024}. \quad (2)$$

We chose DINOv3 ViT-L over the DINOv2-g backbone of CellCLIP because it is about $4\times$ smaller at similar transfer quality, which lets one RTX 5080 encode a full plate in about

TABLE II

CELL PAINTING FLUORESCENCE CHANNELS USED IN MORPHOCLIP.

Ch	Stain	Target
1	MitoTracker / Alexa 647	Mitochondria
2	Phalloidin / Alexa 568	Actin
3	WGA / Alexa 488	Golgi / plasma membrane
4	Concanavalin A / Alexa 488	Endoplasmic reticulum
5	Hoechst 33342	DNA / nucleus

seven minutes. The channel tokens are written to disk once, about 75 MB per plate, and every training run reads from that cache. The backbone never runs during training.

a) *CrossChannelFormer (CCF)*.: A one-layer transformer encoder [18] with 4 heads merges the five channel tokens into one site vector, following the channel-aware design of CellCLIP [9] and Channel ViT [16]. Each token is L2-normalized so the five channels enter at a common scale, then a learned channel embedding $\mathbf{e}_c$ is added so the transformer can tell the stains apart. A learned aggregation token $\mathbf{q}$ is prepended and its output is the site vector:

$$\mathbf{h}_{\text{site}} = [\text{CCF}(\mathbf{q}, \hat{\mathbf{z}}_1 + \mathbf{e}_1, \dots, \hat{\mathbf{z}}_5 + \mathbf{e}_5)]_0, \quad \hat{\mathbf{z}}_c = \mathbf{z}_c / \|\mathbf{z}_c\|_2. \quad (3)$$

The layer is pre-norm with a feed-forward width of 4096 and GELU activations. The transformer can read out joint patterns across stains, for example a change that shows in both the mitochondrial and the ER channel. The CCF holds 12.6M of the 14M trained parameters.

b) *Site-to-well pooling*.: The training unit is a well. Site vectors are averaged under a site mask, so wells with 7, 9 or 16 imaged sites all produce one vector and the order of sites does not matter.

c) *Image projection head*.: A two-layer MLP with LayerNorm, GELU and dropout ($p = 0.3$) maps the well vector $\mathbf{h}_{\text{img}}$ to the shared space:

$$f_{\text{img}}(\mathbf{x}) = \text{L2Norm}(\mathbf{W}_2 \text{Drop}(\text{GELU}(\text{LN}(\mathbf{W}_1 \mathbf{h}_{\text{img}})))), \quad (4)$$

with $\mathbf{W}_1 \in \mathbb{R}^{512 \times 1024}$ and $\mathbf{W}_2 \in \mathbb{R}^{512 \times 512}$.

B. Text Encoder

Each perturbation gets one prompt, built from CPJUMP1 metadata and external annotations with one of four templates:

- **Compound**: “Cell Painting morphological profile of {cell_line} cells treated with the compound {compound_name}. Chemical structure (SMILES): {smiles}. Known target gene: {target_gene}. Target protein function: {gene_function}. Perturbation modality: chemical compound.”
- **CRISPR**: “... with CRISPR-Cas9 knockout of gene {gene_symbol}. Gene description: {gene_description}. Gene function: {gene_function}. Perturbation modality: CRISPR knockout.”
- **ORF**: “... overexpressing gene {gene_symbol} via open reading frame construct. Gene description: {gene_description}. Gene function: {gene_function}. Perturbation modality: ORF overexpression.”

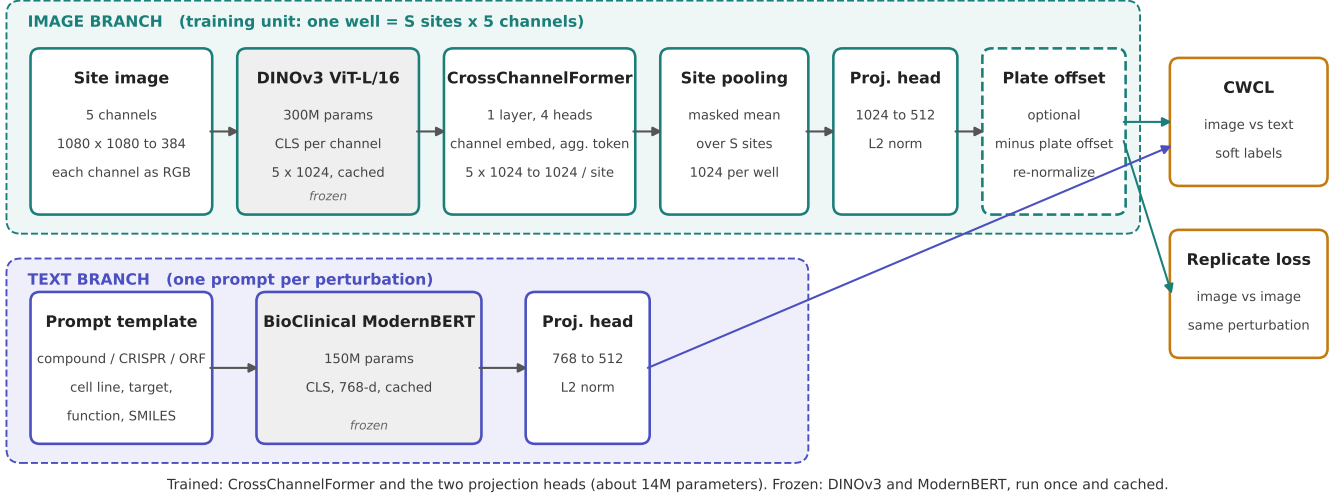


Fig. 1. MorphoCLIP. Each fluorescence channel of a site is encoded on its own by a frozen DINOv3 ViT-L/16, giving five 1024-d CLS tokens. A one-layer CrossChannelFormer attends over the five channel tokens and a learned aggregation token, and the aggregation token’s output is the site vector. Site vectors are mean-pooled into a well vector and projected to the shared 512-d space. The prompt is encoded by a frozen BioClinical ModernBERT and projected to the same space. When plate offsets are on, the image embedding is shifted by its plate’s offset and re-normalized before the losses. Only the CrossChannelFormer and the two projection heads are trained, about 14M parameters.

- **Negative control:** “... treated with DMSO vehicle control. No active perturbation applied.”

Missing fields become “unknown”. Compound annotations come from ChEMBL and the CPJUMP1 metadata. Gene descriptions and functions come from UniProt. Prompts are keyed by the perturbation identifier (broad_sample), so replicate wells of one perturbation share one text vector.

We encode each prompt with a frozen BioClinical ModernBERT [17] (150M parameters, 768-d CLS token). A trainable projection head with the same shape as the image head maps the CLS token to the shared space:

$$f_{\text{txt}}(t) = \text{L2Norm}(\mathbf{W}_4 \text{Drop}(\text{GELU}(\text{LN}(\mathbf{W}_3 \mathbf{h}_{\text{txt}})))), \quad (5)$$

with $\mathbf{W}_3 \in \mathbb{R}^{512 \times 768}$ and $\mathbf{W}_4 \in \mathbb{R}^{512 \times 512}$. The 768-d BERT outputs are cached separately from the projected 512-d vectors, so a change to the projection head never re-runs BERT.

C. Training Objective

a) *Continuously weighted contrastive loss (CWCL).*: InfoNCE [19] treats every off-diagonal pair in a batch as a negative. That is wrong when two wells in the batch carry the same perturbation, and questionable when two compounds share a target. Following CellCLIP [9], we use a soft-label variant. For a batch of N wells with logits $\mathbf{S} = \tau \mathbf{F}_{\text{img}} \mathbf{F}_{\text{txt}}^\top$, an affinity matrix $\mathbf{W} \in [0, 1]^{N \times N}$ sets $w_{ij} = 1$ when wells i and j carry the same perturbation, $w_{ij} = \alpha$ when their perturbations differ but their target-gene sets overlap, and $w_{ij} = 0$ otherwise. The loss is

$$\mathcal{L}_{\text{CWCL}} = -\frac{1}{2N} \sum_{i=1}^N \sum_{j=1}^N \left(\tilde{w}_{ij} \log p_{ij} + \tilde{w}_{ij}^\top \log p_{ij}^\top \right), \quad (6)$$

where p_{ij} is the softmax of row i of $\mathbf{S}$, $\tilde{w}_{ij} = w_{ij} / \sum_k w_{ik}$, and the transposed terms use $\mathbf{S}^\top$ and $\mathbf{W}^\top$ normalized over their own rows. The gene weight α is a hyperparameter; the base model uses $\alpha = 0$ (binary labels) and the soft-label ablation uses $\alpha = 0.6$. The scale τ is learned through its logarithm, which starts at $\ln(1/0.07)$ and is clamped to $[0, \ln 100]$ after every step.

b) *Replicate alignment loss.*: The text loss never asks two replicate wells of one perturbation to be near each other. It only asks each to be near their shared text. We add an image-image term that does. With P_i the set of other wells in the batch that carry the same perturbation as well i and $\mathbf{G} = \tau \mathbf{F}_{\text{img}} \mathbf{F}_{\text{img}}^\top$ with the diagonal masked out,

$$\mathcal{L}_{\text{rep}} = -\frac{1}{|\{i : P_i \neq \emptyset\}|} \sum_{i: P_i \neq \emptyset} \frac{1}{|P_i|} \sum_{j \in P_i} \log \frac{\exp G_{ij}}{\sum_{k \neq i} \exp G_{ik}}. \quad (7)$$

Wells with no in-batch replicate are dropped from the mean, and the term shares the learned temperature with the text loss. The training loss is $\mathcal{L} = \mathcal{L}_{\text{CWCL}} + \lambda_{\text{rep}} \mathcal{L}_{\text{rep}}$; the base model uses $\lambda_{\text{rep}} = 0$ and the ablation uses 0.3. Validation loss is the text term alone, so model selection is not influenced by the added term.

c) *Perturbation-aware batches.*: For the replicate term to have positives, replicate wells must share a batch. Our sampler groups the training wells by perturbation, cuts each group into chunks of two wells, shuffles the chunks and packs them into batches of 256. A repair pass then swaps chunks so that every batch spans at least two plates where that is possible, which keeps in-batch negatives from being confounded with plate. Every ablation run uses this sampler; the base model uses uniform random batches.

d) *Plate offsets (cross-well alignment).*: Batch effects are the main confound in Cell Painting, but in CPJUMP1 a plate is close to synonymous with an experimental condition (one cell line, one perturbation modality, one timepoint). Subtracting a plate’s mean embedding would therefore also subtract the condition, which is exactly what the prompt describes. Our correction is relative to replicate plates instead. At the start of each epoch a gradient-free pass over the training wells computes each plate’s mean embedding $\bar{\mathbf{f}}_p$, and every well is shifted by its plate’s offset from the mean of the plates in the same condition:

$$\tilde{\mathbf{f}}_w = \frac{\mathbf{f}_w - \delta_{p(w)}}{\|\mathbf{f}_w - \delta_{p(w)}\|_2}, \quad \delta_p = \bar{\mathbf{f}}_p - \frac{1}{|C(p)|} \sum_{q \in C(p)} \bar{\mathbf{f}}_q, \quad (8)$$

where $p(w)$ is the plate of well w and $C(p)$ is the set of plates that share plate p ’s condition. Offsets within a condition sum to zero, so the step removes replicate-to-replicate drift and cannot remove the condition itself. The condition of each plate comes from the CPJUMP1 reference metadata; the eleven plates that the reference table does not cover (non-standard seeding density, antibiotics, or compounds on the Cas9 line) are keyed on every condition axis from the experiment table so they are not merged into a standard group. Offsets are constants (no gradient flows through them), apply to image embeddings only, are frozen within an epoch, and are stored in the checkpoint so evaluation and profile export apply the same correction as training. An earlier version of this component subtracted the in-batch plate mean instead. It removed the condition signal and cut retrieval to near chance; Section VI reports that run for completeness.

D. Optimization

We train with AdamW [20] at learning rate 10^{-4} , weight decay 0.2 on weight matrices and none on biases, norms and the temperature, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$. The learning rate warms up linearly for 100 steps and then follows a cosine to zero. We use FP16 mixed precision, gradient clipping at norm 1.0 and a batch of 256 wells. The base model trained on a 100-epoch schedule and its best validation loss came at epoch 9; the ablation runs use a 30-epoch schedule with early stopping after 8 epochs without improvement. With the cache held in RAM an epoch of 55 optimizer steps takes about four seconds on the RTX 5080, so a 30-epoch ablation run finishes in under two minutes of GPU time. Loading the cache into memory takes longer than the training that follows.

V. EXPERIMENTAL SETUP

A. Dataset

We use the CPJUMP1 pilot [5], a public dataset from the JUMP Cell Painting Consortium hosted on the Cell Painting Gallery (S3 prefix `cpg0000-jump-pilot`). It has 51 plates and 3.31 TiB of raw 16-bit TIFF images. The perturbations are 303 small-molecule compounds, 160 CRISPR knockouts and 176 ORF overexpressions, in two cell lines (U2OS and A549), mostly at two timepoints per modality. Each gene target is linked to at least two compounds, which is what

makes compound–gene matching testable. Each plate has 384 wells with 7 to 16 imaged sites per well, and each site has five fluorescence channels at 1080×1080 pixels plus three brightfield channels that we do not use.

B. Data Pipeline

The raw plates are 58 to 118 GiB each, so we stream, encode and delete. Each plate is downloaded to scratch, passed once through the frozen DINOv3 backbone, and removed once the per-channel CLS tokens are on disk. The cache for all 51 plates is about 3.8 GB, or 75 MB per plate.¹ Encoding takes about seven minutes per plate on an RTX 5080, about six hours in total. Text embeddings are cached the same way: each prompt is encoded once by the frozen ModernBERT and the 768-d CLS token stored.

C. Splits

Wells are split by perturbation. Each `broad_sample` is hashed with a fixed seed into train, validation or test at 80/10/10, so all replicate wells of a perturbation land in one split and every split contains all three modalities. Control wells are excluded from training and evaluation. The validation split has 2,220 wells from 98 perturbations (42 compound, 34 CRISPR, 22 ORF), about 24 replicate wells each. The test split has 1,860 wells from 86 perturbations. The hash is deliberately stable: changing it would move every well to a different split and invalidate every split saved on disk.

D. Retrieval Protocol

Replicate wells share one text vector, so the text side of a split has P distinct vectors (98 on validation) and the image side has N wells (2,220). We report four directions, each with an analytic random baseline computed from the same P and N .

- **Well-level image-to-text.** Each of the N wells ranks the P unique texts. One candidate is correct. Random Recall@ k is k/P .
- **Well-level text-to-image.** Each of the P texts ranks all N wells and is scored on the rank of its first correct replicate. With m replicate wells among N candidates, random Recall@ k is $1 - \binom{N-m}{k} / \binom{N}{k}$.
- **Perturbation-level, both directions.** The well embeddings of each perturbation are averaged and re-normalized into one profile, and the P profiles are ranked against the P texts. This is the unit of the standard CPJUMP1 protocol. Random Recall@ k is k/P .

We report Recall@{1, 5, 10} and the median rank of the first correct match. Model selection uses the validation text loss, and we report both the validation and the test split.

An earlier version of this evaluation ranked each well against all N well-texts instead of the P unique texts. Because the pool then held about 24 exact copies of every text vector,

¹An earlier version of the extractor saved each site tensor as a slice of a larger buffer, and `torch.save` wrote the whole buffer each time. The cache was $44\times$ larger than it needed to be. Numbers in earlier drafts of this work quoted that size.

a model that scores identical vectors identically placed all copies at consecutive ranks, which capped Recall@5 and Recall@10 near Recall@1 and made image-to-text retrieval look worse than a shuffled baseline that interleaved the copies. Section VI-E gives the arithmetic. Every number in this paper uses the unique-text pool.

E. Standard CPJUMP1 Benchmark

We also run the benchmark of Chandrasekaran et al. [5] on exported well embeddings, with the pinned copairs implementation the reference code uses. Trained checkpoints encode every well of a plate (controls included) in fp32, subtract the plate’s negative-control mean, and write one profile CSV per plate in the layout the benchmark reads. The benchmark covers the 40 plates at standard seeding density without antibiotics, at two timepoints per modality (short: compound 24 h, ORF 48 h, CRISPR 96 h; long: 48 h, 96 h, 144 h). It measures three things with mean average precision (mAP) against a permutation null: *replicability*, whether replicate wells of one perturbation retrieve each other; *target matching*, whether perturbations of one modality that share a target retrieve each other; and *gene-compound matching*, whether compounds retrieve the CRISPR or ORF perturbations of their target. Each track reports mean mAP and the fraction of perturbations whose mAP passes $q < 0.05$ (fraction retrieved).

One caveat applies to every fraction-retrieved value in this paper and in the reference results: the permutation null is random, so the set of perturbations that pass the threshold changes from run to run of identical code. Only replicability mAP is deterministic. Small differences in fraction retrieved should not be read as effects.

F. Baselines

- **CellProfiler** [6]. Handcrafted features. On CPJUMP1 sister-perturbation matching the published fraction retrieved is 4.3 to 25.1% (mean about 13%) [5]. We did not rerun the CellProfiler pipeline.
- **CellCLIP** [9]. We quote the published Bray-dataset retrieval (best Recall@10 7.4%) and CORUM recall (0.354). We also ran the released CellCLIP checkpoint through our benchmark harness on the short timeline, exporting profiles with CellCLIP’s own encoder path. Those numbers are the closest like-for-like comparison we have, with two caveats noted where they appear: CellCLIP’s evaluation recipe includes a control-fitted KernelPCA batch correction that our export does not apply, and the two are separate runs of a non-deterministic harness.
- **CWA-MSN** [10]. CORUM gene-gene recall 0.386 with 22M parameters. Its benchmarks (CORUM, HuMAP, Reactome) are not among ours, so no comparable number exists.

G. Ablation Design

The base model was trained with binary CWCL labels, uniform random batches, no replicate term and no plate offsets,

on the 100-epoch schedule. The ablations share one config that differs from the base in three ways: perturbation-aware batches, a 30-epoch schedule and early stopping (patience 8). From that shared control we change one thing at a time: gene-aware soft labels ($\alpha = 0.6$), the replicate alignment loss ($\lambda_{\text{rep}} = 0.3$), and plate offsets. The plate-offset variant was run three times: once with the earlier in-batch mean subtraction, once with condition-relative offsets for the 40 plates the reference metadata covers and a zero offset for the other 11, and once with offsets for all 51 plates. A final run turns all three additions on together. Every run uses seed 42, batch size 256 and the same split. All retrieval numbers below therefore come from one seed. Where two runs differ by a few points at 98 or 86 candidates, one perturbation is worth about one point, and we do not read such differences as effects.

H. Implementation

Everything runs on one RTX 5080 (16 GB). AdamW, learning rate 10^{-4} , weight decay 0.2, 100 warm-up steps then cosine decay, FP16, gradient clipping at 1.0. CCF: one layer, four heads. Projection heads: hidden width 512, dropout 0.3. The code, configs and split manifests are released with the paper.

VI. RESULTS AND DISCUSSION

All numbers come from a single seed. Retrieval numbers are on the validation and test splits described in Section V-D; benchmark numbers are on the 40 benchmark-eligible plates. We give the random baseline next to every retrieval figure and note where a difference is smaller than one perturbation’s worth of recall.

A. Cross-Modal Retrieval

Table III reports the base model on both splits. On validation, ranking the 98 perturbation profiles against the 98 texts gives Recall@10 of 38.8% image-to-text and 39.8% text-to-image, against a random 10.2%, with a median rank of 13. On the 86 test perturbations the same checkpoint gives 37.2% and 44.2% against a random 11.6%. The two directions are within a few perturbations of each other on both splits, and both sit close to four times chance.

Well-level text-to-image is the one direction that looks weak in Recall@10 (23.5% against 9.7%), but that metric asks the text to place one of its 24 replicate wells in the top ten of 2,220, and its median rank of 30 says the first replicate is usually found within the top 1.4% of the pool. Once wells are averaged into profiles the two directions agree.

B. Ablations

Table IV lists the single-factor runs. Three points come out of it.

First, the retrieval effects of the three additions are within the noise of one seed. On validation the replicate loss adds 7 points of image-to-text recall over the control and soft labels add 5 points of text-to-image recall, but on the test split the control scores highest on image-to-text and the ordering of the

TABLE III

RETRIEVAL OF THE BASE MODEL ON THE VALIDATION SPLIT (2,220 WELLS, 98 PERTURBATIONS) AND THE TEST SPLIT (1,860 WELLS, 86 PERTURBATIONS). WELL-LEVEL IMAGE-TO-TEXT RANKS THE UNIQUE TEXTS; WELL-LEVEL TEXT-TO-IMAGE RANKS ALL WELLS AND SCORES THE FIRST CORRECT REPLICATE; PERTURBATION-LEVEL RANKS MEAN PROFILES AGAINST TEXTS. RANDOM ROWS ARE ANALYTIC. MEDIAN RANK IS OVER THE CANDIDATE POOL OF EACH DIRECTION (P TEXTS, N WELLS, OR P PROFILES).

Split	Direction	R@1	R@5	R@10	Med. rank
Val	well image $\rightarrow$ text	5.9	23.3	39.4	14
	well text $\rightarrow$ image	4.1	15.3	23.5	30
	pert. image $\rightarrow$ text	4.1	20.4	38.8	13
	pert. text $\rightarrow$ image	5.1	18.4	39.8	13
	random, single positive	1.0	5.1	10.2	49.5
	random, well text $\rightarrow$ image	1.0	5.0	9.7	99
Test	well image $\rightarrow$ text	5.5	22.0	41.4	13
	well text $\rightarrow$ image	4.7	18.6	26.7	25
	pert. image $\rightarrow$ text	4.7	17.4	37.2	14
	pert. text $\rightarrow$ image	8.1	23.3	44.2	13
	random, single positive	1.2	5.8	11.6	43.5
	random, well text $\rightarrow$ image	1.2	5.7	11.0	86

four runs changes. A difference of 5 points at 98 candidates is five perturbations, and the same checkpoint moves by more than that between the two splits. We therefore do not claim a retrieval gain for any single addition on this evidence. The replicate loss does converge faster: its best epoch is 7 against 11 for the control.

Second, the in-batch version of the plate offset destroyed the model. Subtracting the mean of whichever wells of a plate happened to share a batch cut perturbation-level retrieval to 15% against a 10% floor, and the model stopped improving after four epochs. Evaluating that checkpoint with the correction switched off at inference did not help (12%), so the training itself was harmed. The mechanism is the one Section V-G anticipates: in CPJUMP1 a plate is nearly one condition, and the plate mean carries the modality and cell line that the prompt names. The condition-relative offset removes that failure. With offsets for all 51 plates the run trains normally, its validation loss matches the control, and its retrieval is at control level. Extending the condition map from 40 to all 51 plates moved image-to-text from 36.7 to 41.8 on validation, which is five perturbations and again within noise.

Third, turning all three on together gives 42.9 and 43.9 on validation and 46.5 and 45.3 on test, with the lowest validation text loss of the campaign (5.396 against the control’s 5.438) at its best epoch. Those figures are among the best in the table on both splits, but they are not additive: the combined run does not reach the replicate-loss run’s 44.9 on validation image-to-text or the control’s 50.0 on test. On this evidence the three additions are compatible and none of them hurts, and that is as far as retrieval alone lets us go.

C. Standard CPJUMP1 Benchmark

Table V reports fraction retrieved and mean mAP on the standard benchmark for the base model, the replicate-loss run, the plate-offset run and the combined run.

a) Replicability.: The base model replicates compounds well (fraction retrieved 0.30 to 0.51, mAP 0.34 to 0.53), CRISPR less well (0.03 to 0.18) and ORF hardly at all (below 0.05). The replicate loss raises fraction retrieved on all twelve tracks and mAP on all twelve, from a mean of 0.178 to 0.248 and 0.298 to 0.343. This is the one effect in the paper that is consistent across every track and both metrics, and mAP is the deterministic one. It is also partly definitional: the loss pulls replicate wells of one perturbation together, and replicability measures whether replicate wells retrieve each other. The gain says the loss does what it was designed to do. It does not by itself say the embedding is more biologically informative. Plate offsets on their own leave replicability where the control has it (mean mAP 0.292 against 0.298), so the correction that was meant to help cross-plate matching does not show up in the cross-plate metric. The combined run has the highest replicability of the campaign, mean mAP 0.369 and fraction retrieved 0.277, and its mAP is above the replicate-loss run on ten of the twelve tracks and level on the two ORF tracks it does not beat. On this benchmark the replicate loss is doing the work and the other two additions do not undo it.

b) Target matching.: Mean fraction retrieved falls from 0.339 for the base model to 0.262 with the replicate loss and 0.172 for the combined run, but the population being scored is not the same. The benchmark only admits perturbations that passed the replicability gate into target matching, so the base model is scored on 536 consensus profiles, the replicate-loss model on 748 and the combined model on 856. The added profiles are the ones that were only just replicable, and they lower the mean. The CRISPR tracks have between 2 and 30 targets each (the U2OS short track has two or three), so their fractions move by whole steps and should not be read at all; the plate-offset run’s 0.420 mean is two CRISPR tracks at 1.0 with a handful of targets each. On the compound tracks, which have 94 to 250 targets, mean mAP runs from 0.13 to 0.28 for the base model and 0.11 to 0.21 for the replicate-loss run, and no variant is consistently ahead.

c) Gene-compound matching.: At or near zero for every variant we ran. Mean fraction retrieved over the admitted pairs is 0.005 for the base model (2 of 219 pairs pass the gate), 0.000 with the replicate loss (0 of 385), 0.010 with plate offsets (3 of 179) and 0.029 for the combined run (3 of 480, and that figure is one compound-ORF pair at 0.4 with five targets in it). The compound-CRISPR pairs, which have 16 to 33 targets, reach 0.06 at most, and no pair is nonzero in more than two of the four runs. Mean mAP on the admitted pairs is 0.08 to 0.14. The plate-offset run is the only one whose nonzero pairs are all compound-CRISPR, and its gene-compound mAP (0.140) is the highest of the four; that is the direction a cross-plate correction should push, but three pairs at one seed is not a result. Cross-modality matching is the task this line of work is aimed at, and none of the training changes moved it beyond that. We report it as an open failure and return to it in Section VII.

TABLE IV

SINGLE-FACTOR ABLATIONS FROM A SHARED CONTROL (PERTURBATION-AWARE BATCHES, 30-EPOCH SCHEDULE, EARLY STOPPING, SEED 42). PERTURBATION-LEVEL RECALL@10 IN PERCENT, MEDIAN RANK OVER P TEXTS, ON VALIDATION (98 PERTURBATIONS) AND TEST (86). RANDOM RECALL@10 IS 10.2 ON VALIDATION AND 11.6 ON TEST. ONE PERTURBATION IS ABOUT ONE POINT. BEST EPOCH IS THE EPOCH SELECTED BY VALIDATION TEXT LOSS.

Run	Change from control	Validation			Test			Best ep.
		i→t	t→i	Med.	i→t	t→i	Med.	
Base model	100-epoch schedule, random batches	38.8	39.8	13	37.2	44.2	14	9
Control	none	37.8	37.8	13	50.0	41.9	10	11
Soft labels	$\alpha = 0.6$	37.8	42.9	14	44.2	47.7	12	13
Replicate loss	$\lambda_{\text{rep}} = 0.3$	44.9	41.8	13	47.7	46.5	12	7
Plate offsets (in-batch, removed)	use_cwa	15.3	10.2	45	—	—	—	4
Plate offsets (40 of 51 plates)	use_cwa	36.7	39.8	13	—	—	—	8
Plate offsets (all 51 plates)	use_cwa	41.8	38.8	13	41.9	43.0	12	7
All three	soft + replicate + offsets	42.9	43.9	13	46.5	45.3	11	9

TABLE V

STANDARD CPJUMP1 BENCHMARK ON THE 40 ELIGIBLE PLATES. REPLICABILITY PER MODALITY, CELL LINE AND TIMEPOINT: FRACTION RETRIEVED AT $q < 0.05$ WITH MEAN MAP IN PARENTHESES. TARGET MATCHING AND GENE-COMPOUND MATCHING ARE MEANS OVER THEIR TRACKS. FRACTION RETRIEVED VARIES BETWEEN RUNS OF IDENTICAL CODE; ONLY REPLICABILITY MAP IS DETERMINISTIC.

Track		Base	Replicate loss	Plate offsets	All three
Compound A549	short	0.353 (0.392)	0.435 (0.458)	0.327 (0.384)	0.497 (0.493)
	long	0.513 (0.526)	0.650 (0.583)	0.503 (0.513)	0.703 (0.626)
Compound U2OS	short	0.297 (0.344)	0.379 (0.404)	0.310 (0.351)	0.422 (0.439)
	long	0.369 (0.413)	0.471 (0.474)	0.395 (0.421)	0.503 (0.496)
CRISPR A549	short	0.173 (0.286)	0.271 (0.343)	0.092 (0.239)	0.340 (0.375)
	long	0.183 (0.296)	0.376 (0.356)	0.134 (0.282)	0.366 (0.385)
CRISPR U2OS	short	0.026 (0.175)	0.049 (0.207)	0.052 (0.182)	0.078 (0.232)
	long	0.170 (0.277)	0.219 (0.290)	0.160 (0.259)	0.258 (0.318)
ORF A549	short	0.006 (0.136)	0.012 (0.158)	0.012 (0.138)	0.019 (0.174)
	long	0.000 (0.128)	0.006 (0.142)	0.000 (0.126)	0.006 (0.141)
ORF U2OS	short	0.043 (0.171)	0.099 (0.223)	0.062 (0.209)	0.099 (0.221)
	long	0.006 (0.133)	0.012 (0.146)	0.012 (0.140)	0.031 (0.160)
Replicability, mean of 12		0.178 (0.298)	0.248 (0.343)	0.172 (0.292)	0.277 (0.369)
Target matching, mean of 8		0.339 (0.197)	0.262 (0.160)	0.420 (0.194)	0.172 (0.148)
Gene-compound, mean of pairs		0.005 (0.123)	0.000 (0.082)	0.010 (0.140)	0.029 (0.082)

D. Comparison with Baselines

a) *CellProfiler*.: The published CellProfiler fraction retrieved on CPJUMP1 sister-perturbation matching is 4.3 to 25.1% [5]. That is a target-matching figure, so the like-for-like rows in Table V are the target-matching tracks, not replicability. There the base model’s compound tracks span 0.07 to 0.31 (mean 0.16), inside the published range, and its CRISPR tracks are too small to read. We did not rerun CellProfiler on our harness, so this is our run against the published one, and we do not claim an advantage over CellProfiler from it. Replicability, the task on which our numbers are highest, is easier than target matching and is not what the published range measures.

b) *CellCLIP*.: Table VI sets our short-timeline replicability against the released CellCLIP checkpoint run through the same harness. CellCLIP’s evaluation recipe fits a KernelPCA on control wells before scoring; ours applies no such step, and the two rows are separate runs of the harness. On compounds the base model is close to CellCLIP (0.353 and 0.297 against 0.366 and 0.353) and the replicate-loss and combined runs are

above it (0.435 and 0.379; 0.497 and 0.422). On CRISPR the picture splits by cell line: MorphoCLIP is well ahead on A549 (0.173 to 0.340 against 0.020) and behind on U2OS (0.026 to 0.078 against 0.118), and CellCLIP is ahead on U2OS ORF (0.119 against 0.043 to 0.099). Both models are at or near zero on gene-compound matching. Given the harness variance and the batch-correction difference, we read this as parity or better on compounds with a much smaller model, and as no evidence either way on the genetic tracks.

c) *CWA-MSN*.: CWA-MSN reports CORUM recall of 0.386 on RxRx3-core. We have no number on that benchmark and make no comparison. Its core mechanism, pulling embeddings of one perturbation together across wells, is closer to our replicate alignment loss than to our plate offsets, and it is that mechanism, not the offsets, that moved replicability here.

E. Evaluation Pitfall: Duplicated Candidates

The first version of our evaluation ranked each well against all 2,220 well-texts. Replicate wells share one text vector, so that pool contained 98 distinct vectors, each present about 24 times. A model scores identical vectors identically, so the

TABLE VI
SHORT-TIMELINE REPLICABILITY, FRACTION RETRIEVED, SAME BENCHMARK HARNESS. CELLCLIP ROW USES THE RELEASED CHECKPOINT AND ITS OWN EXPORT PATH. SEPARATE RUNS OF A NON-DETERMINISTIC HARNESS.

Track	CellCLIP	Base	+ replicate loss	All three
Compound A549	0.366	0.353	0.435	0.497
Compound U2OS	0.353	0.297	0.379	0.422
CRISPR A549	0.020	0.173	0.271	0.340
CRISPR U2OS	0.118	0.026	0.049	0.078
ORF U2OS	0.119	0.043	0.099	0.099

copies of the correct text always occupy a run of consecutive ranks. If the correct text ranks first among the 98 distinct texts, its copies fill ranks 1 to 24, and Recall@5 and Recall@10 collapse to Recall@1. Against that pool the base model scored Recall@10 of 7.1% while a random ranking, which interleaves the copies, scored 10.4%, and we concluded that image-to-text retrieval was below chance. The observed median rank of 289 is what the arithmetic predicts for a median unique rank of about 13 with 24 copies each ($12 \times 24 + 1$). Ranking the 98 unique texts instead gives the 39.4% in Table III. Text-to-image did not have this problem, because its candidates are the wells, which are all distinct. The mask of positives is symmetric between the two directions; the multiplicity of candidates is not, and the random baseline has to follow the candidates.

F. Cached Feature Analysis

We checked that channel identity survives the frozen backbone. On plate BR00116991 (500 sites) a variance decomposition of the cached tokens attributes 31.1% of variance to differences between channels, 37.1% to differences between sites and 31.8% to the residual (Figure 2). Same-channel tokens from different sites have mean cosine similarity 0.84; different-channel tokens from the same site have 0.75, and the gap holds for every channel pair (Figure 3). The DNA channel is the most distinct at 0.73 to the other four. A PCA of the 5×1024 space reaches 90% of variance at 23 components (Figure 4). Channel structure is present in the input, which is what the CCF’s channel embedding is there to use.

G. Cost

The DINOv3 forward pass is the expensive step and it runs once: about seven minutes per plate, six hours for the dataset. After that the cost of training is dominated by reading the 3.8 GB cache into memory. With the cache in RAM, an epoch of 55 steps at batch size 256 takes about four seconds on the RTX 5080 and a few gigabytes of GPU memory, so a full ablation run is a matter of minutes and the whole campaign in this paper fits in an afternoon on one card. Both backbones stay frozen; the CrossChannelFormer and the two projection heads are the only trained parts.

VII. CONCLUSION

MorphoCLIP maps Cell Painting wells and text descriptions of perturbations into one 512-dimensional space with about

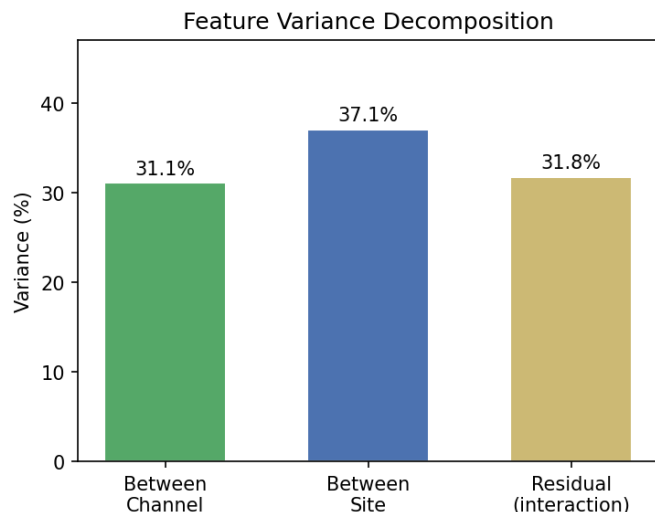


Fig. 2. Variance decomposition of the cached DINOv3 tokens on plate BR00116991: between-channel, between-site and residual components.

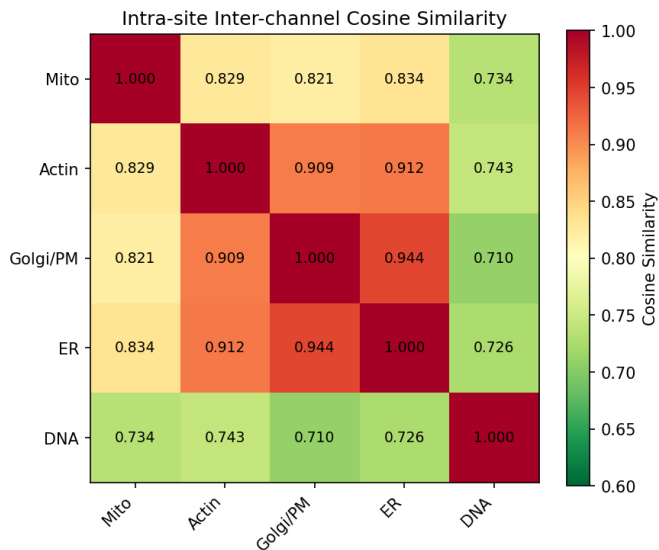


Fig. 3. Mean cosine similarity between fluorescence channels in the cached token space. The DNA channel is the most distinct.

14M trained parameters on top of two frozen backbones. On CPJUMP1 it retrieves the right text for a held-out perturbation profile, or the right profile for a text, within the top ten of 98 candidates about 39% of the time on validation and 37 to 44% on test, against a chance rate of 10 to 12%. One model handles compounds, CRISPR knockouts and ORF overexpressions. On the standard benchmark its compound replicability is level with or above the released CellCLIP checkpoint on the same harness, at one hundredth of that model’s size, and its compound target matching sits inside the published CellProfiler range.

The ablations are more modest than we hoped when we designed them. Gene-aware soft labels, the replicate alignment

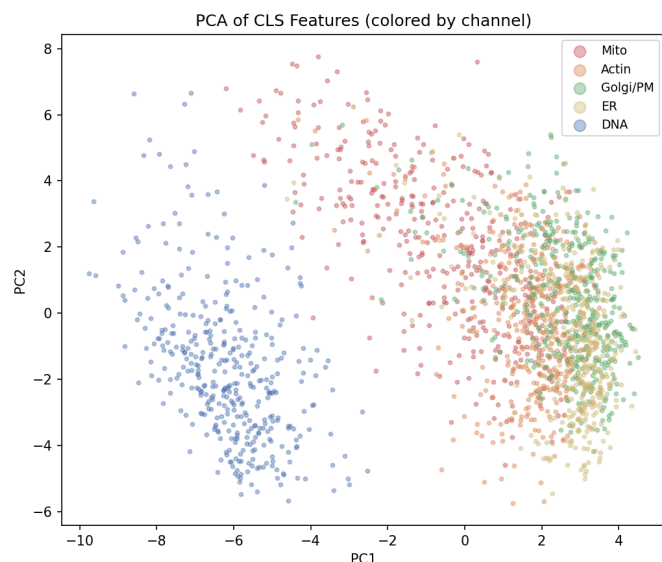


Fig. 4. Cumulative explained variance of a PCA over the 5×1024 -d token space; 90% is reached at 23 components.

loss and condition-relative plate offsets each change validation retrieval by a few perturbations, and the test split reorders them. What survives is that the replicate loss raises benchmark replicability on every track and in the deterministic metric, and that the naive in-batch plate correction is harmful for a reason we can now state: on CPJUMP1 the plate mean carries the condition, and the prompt describes the condition. The condition-relative offset removes that failure without, so far, buying anything measurable at retrieval or on the benchmark. With all three additions on, replicability is the highest we measured (mean mAP 0.37 against 0.30 for the base model), and retrieval is among the best in the campaign without being additive.

Cross-modality gene–compound matching, the task this line of work exists for, is at or near zero for every model we trained: at most three compound–gene pairs pass the significance gate in any run, and we have no positive result on it.

A. Limitations and Future Work

Every retrieval number in this paper is from one seed, and the split-to-split movement of a single checkpoint is as large as any ablation effect. Seeds are the first thing to add, for the control and for the replicate-loss and combined runs. Fraction retrieved on the standard benchmark varies between runs of the same code, so those tables should be read on the mAP columns and treated as indicative on the others. We did not rerun CellProfiler on our harness, and the CellCLIP row comes from a separate run with a different export path and, in its own recipe, a batch-correction step ours lacks. The replicability gain from the replicate loss is in part the loss doing what it says, and a claim that the embedding is more informative needs a task the loss does not optimize, which target matching did not deliver at this seed count.

On cross-modality, one concrete lead is the choice of gene annotation. Training soft labels use each compound’s primary target (160 distinct genes) while the benchmark’s gene–compound track uses the full target list (758 distinct genes, off-targets included). Aligning the two, or training on the wider list, is a one-line change we have not measured. A second lead is prompt content: whether the SMILES string and gene-function text carry signal through BioClinical ModernBERT, or are inert, can be tested by ablating them out of the templates. Beyond CPJUMP1, the full JUMP corpus of about 140,000 perturbations across many sites and cell lines is the setting in which a text-supervised model would have to prove that its embeddings transfer.

We release the code, configs, split manifests and the evaluation harness with the paper.

REFERENCES

- [1] O. J. Wouters, M. McKee, and J. Luyten, “Estimated research and development investment needed to bring a new medicine to market, 2009–2018,” *JAMA*, vol. 323, no. 9, pp. 844–853, 2020.
- [2] D. Sun, W. Gao, H. Hu, and S. Zhou, “Why 90% of clinical drug development fails and how to improve it?” *Acta Pharmaceutica Sinica B*, vol. 12, no. 7, pp. 3049–3062, 2022.
- [3] M.-A. Bray, S. Singh, H. Long, and A. E. Carpenter, “Cell painting, a high-content image-based assay for morphological profiling using multiplexed fluorescent dyes,” *Nature Protocols*, vol. 11, no. 9, pp. 1757–1774, 2016.
- [4] S. Seal *et al.*, “Cell painting: A decade of discovery and innovation in cellular imaging,” *Nature Methods*, 2024.
- [5] S. N. Chandrasekaran, J. Ackerman, E. Alix, D. M. Ando, J. Arevalo, M. Berber, N. Beresford, L. Bornholdt, B. Bostock *et al.*, “JUMP cell painting dataset: morphological impact of 136,000 chemical and genetic perturbations,” *Nature Methods*, vol. 21, pp. 1114–1121, 2024.
- [6] A. E. Carpenter, T. R. Jones, M. R. Lamprecht, C. Clarke, I. H. Kang, O. Friman, D. A. Guertin, J. H. Chang, R. A. Lindquist, J. Moffat *et al.*, “CellProfiler: image analysis software for identifying and quantifying cell phenotypes,” *Genome Biology*, vol. 7, no. 10, p. R100, 2006.
- [7] A. Sánchez-Fernández, T. Langer, G. Ecker *et al.*, “CLOOME: Contrastive learning unlocks bioimaging databases for queries with chemical structures,” *Nature Communications*, 2023.
- [8] P. Fradkin *et al.*, “How molecules impact cells: Unlocking contrastive PhenoMolecular retrieval,” in *Advances in Neural Information Processing Systems*, 2024.
- [9] M. Lu, E. Weinberger, C. Kim, and S.-I. Lee, “CellCLIP: Learning perturbation effects in cell painting via text-guided contrastive learning,” in *Advances in Neural Information Processing Systems*, 2025.
- [10] P.-J. Huang, Y.-H. Liao, S. Kim, N. Park, J. Park, and D. Shin, “Efficient cell painting image representation learning via cross-well aligned masked siamese network,” *arXiv preprint arXiv:2509.19896*, 2025.
- [11] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International Conference on Machine Learning*. PMLR, 2021, pp. 8748–8763.
- [12] O. Simeoni *et al.*, “DINOv3: Self-supervised learning via gram anchoring,” *arXiv preprint arXiv:2508.10104*, 2025.
- [13] N. Moshkov *et al.*, “Self-supervision advances morphological profiling by unlocking powerful image representations,” *Scientific Reports*, 2025.
- [14] V. Ljosa, K. L. Sokolnicki, and A. E. Carpenter, “Annotated high-throughput microscopy image sets for validation,” *Nature Methods*, vol. 9, no. 7, pp. 637–637, 2012.
- [15] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby *et al.*, “DINOv2: Learning robust visual features without supervision,” *Transactions on Machine Learning Research*, 2024.
- [16] Y. Bao, S. Sivanandan, and T. Karaletsos, “Channel vision transformers: An image is worth $1 \times 16 \times 16$ words,” in *International Conference on Learning Representations*, 2024.

- [17] T. Sounack *et al.*, “BioClinical ModernBERT: A state-of-the-art long-context encoder for biomedical and clinical NLP,” *arXiv preprint arXiv:2506.10896*, 2025.
- [18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [19] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” in *Advances in Neural Information Processing Systems*, 2018.
- [20] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *International Conference on Learning Representations*, 2019.